Conflict Graphs

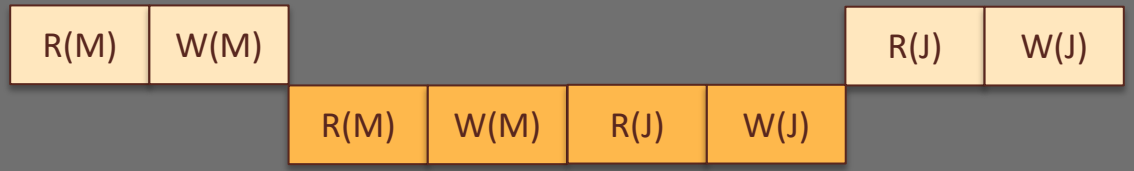
Conflict graph

- Nodes: Transactions T_i
- Edges: $T_i \rightarrow T_j$ whenever schedule has conflicting actions from T_i to T_j.

Conflict equiv.
Serializable



Not conflict equiv.
Non-serializable



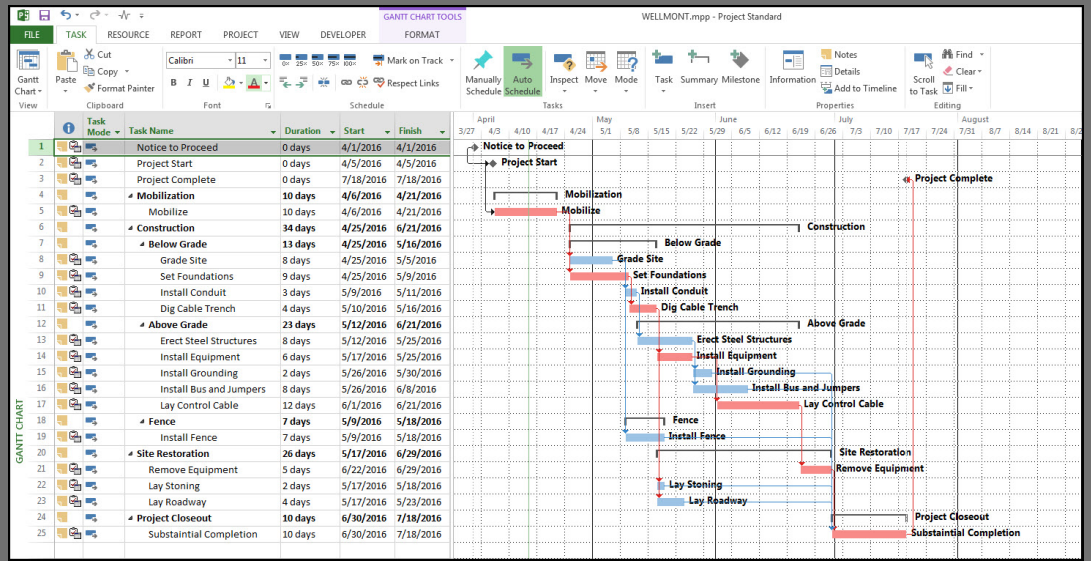
Meaning

Edge $T_i \rightarrow T_j$ = dependence

A serial schedule would schedule T_i before T_j.

Topological order of DAG

Fact: Any DAG can have its nodes put in linear order that respects its directed edges.



Conflict serializability

Schedule is conflict serializable if its conflict graph is acyclic.